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```
MariaDB [(none)]> create database project_db;  
Query OK, 1 row affected (0.030 sec)
```

```
MariaDB [(none)]> use project_db;  
Database changed  
MariaDB [project_db]>
```

--Exercise 7.24 - Creating the Projects Schema

```
MariaDB [project_db]> CREATE TABLE Employee (  
->     empNo VARCHAR(10) PRIMARY KEY,  
->     fName VARCHAR(50),  
->     lName VARCHAR(50),  
->     address VARCHAR(255),  
->     DOB DATE,  
->     sex CHAR(1) CHECK (sex IN ('M', 'F')),  
->     position VARCHAR(50) CHECK (position IN ('Manager', 'Team Leader',  
'Analyst', 'Software Developer')),  
->     deptNo VARCHAR(10)  
-> );  
Query OK, 0 rows affected (0.078 sec)
```

```
MariaDB [project_db]> CREATE TABLE Department (  
->     deptNo VARCHAR(10) PRIMARY KEY,  
->     deptName VARCHAR(100),  
->     mgrEmpNo VARCHAR(10)  
-> );  
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [project_db]> CREATE TABLE Project (  
->     projNo VARCHAR(10) PRIMARY KEY,  
->     projName VARCHAR(100),  
->     deptNo VARCHAR(10),  
->     FOREIGN KEY (deptNo) REFERENCES Department(deptNo)  
-> );  
Query OK, 0 rows affected (0.027 sec)
```

```
MariaDB [project_db]> CREATE TABLE WorksOn (  
->     empNo VARCHAR(10),  
->     projNo VARCHAR(10),
```

```

->     dateWorked DATE,
->     hoursWorked INT CHECK (hoursWorked >= 0 AND hoursWorked <= 40),
->     PRIMARY KEY (empNo, projNo, dateWorked),
->     FOREIGN KEY (empNo) REFERENCES Employee(empNo),
->     FOREIGN KEY (projNo) REFERENCES Project(projNo)
-> );

```

Query OK, 0 rows affected (0.038 sec)

```

--Because SQL executes commands one after the other
--neither table can be created first if they both strictly require the other to
already exist.
-- that is why I am altering the table

```

```

MariaDB [project_db]> ALTER TABLE Employee
-> ADD FOREIGN KEY (deptNo) REFERENCES Department(deptNo);
Query OK, 0 rows affected (0.067 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

```

MariaDB [project_db]> ALTER TABLE Department
-> ADD FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo);
Query OK, 0 rows affected (0.052 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

```

-- Disable FK checks temporarily to insert circular data
MariaDB [project_db]> SET FOREIGN_KEY_CHECKS = 0;
Query OK, 0 rows affected (0.028 sec)

```

```

MariaDB [project_db]> INSERT INTO Employee (empNo, fName, lName, address, DOB, sex,
position, deptNo) VALUES
-> ('E1', 'John', 'Doe', '123 Elm St', '1980-01-01', 'M', 'Team Leader', 'D1'),
-> ('E2', 'Jane', 'Smith', '456 Oak St', '1975-05-15', 'F', 'Manager', 'D1'), --
Female Manager
-> ('E3', 'Bob', 'Jones', '789 Pine St', '1990-08-20', 'M', 'Software
Developer', 'D2'),
-> ('E4', 'Alice', 'Brown', '321 Maple St', '1985-11-30', 'F', 'Manager', 'D2');
-- Female Manager
Query OK, 4 rows affected (0.039 sec)
Records: 4 Duplicates: 0 Warnings: 0

```

```

MariaDB [project_db]> INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES
-> ('D1', 'IT', 'E2'),
-> ('D2', 'HR', 'E4');
Query OK, 2 rows affected (0.029 sec)
Records: 2 Duplicates: 0 Warnings: 0

```

```

MariaDB [project_db]> INSERT INTO Project (projNo, projName, deptNo) VALUES

```

```

-> ('SCCS', 'System Control', 'D1'),
-> ('P2', 'Payroll Upgrade', 'D2');
Query OK, 2 rows affected (0.028 sec)
Records: 2 Duplicates: 0 Warnings: 0

```

```

MariaDB [project_db]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked)
VALUES
-> ('E1', 'SCCS', '2025-10-01', 10),
-> ('E1', 'P2', '2025-10-02', 20),
-> ('E2', 'SCCS', '2025-10-01', 15),
-> ('E3', 'P2', '2025-10-03', 40);
Query OK, 4 rows affected (0.029 sec)
Records: 4 Duplicates: 0 Warnings: 0

```

```

MariaDB [project_db]> SET FOREIGN_KEY_CHECKS = 1;
Query OK, 0 rows affected (0.001 sec)

```

--Exercise 7.25 - Creating the Female Managers View

```

MariaDB [project_db]> CREATE VIEW FemaleManagedProjects AS
-> SELECT p.projNo, p.projName, p.deptNo
-> FROM Project p
-> JOIN Department d ON p.deptNo = d.deptNo
-> JOIN Employee e ON d.mgrEmpNo = e.empNo
-> WHERE e.sex = 'F'
-> ORDER BY p.projNo;

```

Query OK, 0 rows affected (0.043 sec)

-- test view

```

MariaDB [project_db]> SELECT * FROM FemaleManagedProjects;

```

```

+-----+-----+-----+
| projNo | projName      | deptNo |
+-----+-----+-----+
| P2     | Payroll Upgrade | D2     |
| SCCS   | System Control | D1     |
+-----+-----+-----+
2 rows in set (0.042 sec)

```

--Exercise 7.26 - Creating the Employee Project Details View

```

MariaDB [project_db]> CREATE VIEW EmployeeProjectHours AS
-> SELECT e.empNo, e.fName, e.lName, p.projName, w.hoursWorked
-> FROM Employee e
-> JOIN WorksOn w ON e.empNo = w.empNo
-> JOIN Project p ON w.projNo = p.projNo;

```

Query OK, 0 rows affected (0.008 sec)

```
MariaDB [project_db]> SELECT * FROM EmployeeProjectHours;
```

empNo	fName	lName	projName	hoursWorked
E1	John	Doe	Payroll Upgrade	20
E3	Bob	Jones	Payroll Upgrade	40
E1	John	Doe	System Control	10
E2	Jane	Smith	System Control	15

4 rows in set (0.003 sec)

--Exercise 7.27 - Analyzing the EmpProject View

```
MariaDB [project_db]> CREATE VIEW EmpProject(empNo, projNo, totalHours)
-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)
-> FROM Employee e, Project p, WorksOn w
-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo
-> GROUP BY w.empNo, w.projNo;
```

Query OK, 0 rows affected (0.034 sec)

```
MariaDB [project_db]> SELECT * FROM EmpProject;
```

empNo	projNo	totalHours
E1	P2	20
E1	SCCS	10
E2	SCCS	15
E3	P2	40

4 rows in set (0.031 sec)

Analysis:

This query is valid. It returns a summary of all employees, showing the total hours they have aggregated per project.

```
MariaDB [project_db]> SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';
```

projNo
SCCS
SCCS

2 rows in set (0.029 sec)

Analysis:

This query is valid. It filters the view to only show rows where the project number is 'SCCS', returning that project number for every employee who worked on it.

```
MariaDB [project_db]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';
```

```
+-----+
| COUNT(projNo) |
+-----+
|          2 |
+-----+
```

1 row in set (0.002 sec)

Analysis:

This query is valid. It counts the number of distinct projects that employee 'E1' has worked on by counting the rows returned for that employee.

```
MariaDB [project_db]> SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

```
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E1    |          20 |
| E2    |          15 |
| E3    |          40 |
+-----+-----+
```

3 rows in set (0.028 sec)

Analysis:

This query is has some logic error.

It groups by empNo but selects totalHours without an aggregate function (like SUM()).

-- this is to correct the above Query because it is not returning the full hours
-- we need the SUM function.

```
MariaDB [project_db]> SELECT empNo, SUM(totalHours) AS overallTotalHours
-> FROM EmpProject
-> GROUP BY empNo;
```

```
+-----+-----+
| empNo | overallTotalHours |
+-----+-----+
| E1    |          30 |
| E2    |          15 |
| E3    |          40 |
+-----+-----+
```

```
+-----+-----+
3 rows in set (0.002 sec)
```